

**Livelihoods at the Crossroads:
Socioeconomic Impact of the Proposed
Kedarnath Ropeway on
Pilgrimage-Dependent Workers**

Project Report

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Summary. The Kedarnath ropeway project would reduce a multi-hour pilgrimage trek to a short aerial ride. This report assesses who benefits and who bears the cost. A primary survey of 46 workers across five occupational categories finds that opposition is concentrated among animal handlers, whose livelihoods depend entirely on the trail economy, and is driven by high vulnerability rather than information deficits. A synthetic population of 1,000 households is generated and validated against the observed sample, then used to simulate three ropeway scenarios. Under moderate displacement assumptions, animal handlers lose approximately 62 % of their income; under severe assumptions, 12 % of the synthetic population falls below the poverty line. A targeted compensation scheme costing Rs. 3.5–4.5 crore can restore a three-month income buffer for all displaced households — a small fraction of the ropeway project budget.

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1. Introduction and Motivation

The Kedarnath pilgrimage in Uttarakhand is one of India’s most visited religious destinations, drawing several hundred thousand pilgrims each year during the May to November yatra season. The 16-kilometre foot trail from Gaurikund to the Kedarnath temple has, over decades, sustained a dense local labour market: pony and mule handlers who carry pilgrims, porters who carry luggage and the disabled, dhaba and lodge operators who feed and shelter them, and guides and priests who complement the spiritual dimension of the journey.

The state government has proposed an aerial ropeway connecting Gaurikund and Kedarnath (Uttarakhand Tourism Development Board, 2022). The ropeway would reduce travel time from four to six hours on foot to under ten minutes by cable car, improving accessibility and pilgrim comfort. However, most of these improvements accrue to pilgrims, while the costs fall on the workers whose livelihoods depend on foot traffic along the trail.

This project has three objectives. First, to document the livelihood profiles of different occupational groups using a primary survey and a set of composite livelihood indices. Second, to identify the predictors of ropeway support and opposition among workers. Third, to quantify the income and poverty impacts under three displacement scenarios using a synthetic population, and to estimate the minimum fiscal cost of a compensation scheme.

The analysis is grounded in the Sustainable Livelihoods Approach (Chambers and Conway, 1992; DFID, 1999; Ellis, 2000), which frames household welfare in terms of assets, vulnerability, and adaptive capacity rather than income alone. This framing is particularly appropriate for a highly seasonal, mono-dependent economy where income flows are concentrated in a six-month window and off-season fallbacks are scarce.

2. Data Collection

A structured questionnaire survey was administered to pilgrimage-route workers at Kedarnath during the 2024 yatra season. Respondents were purposively sampled across the major occupational categories present at the site, with the aim of covering the full range of trail-dependent livelihoods.

The final sample comprises 46 respondents across five occupational categories (Table 1), covering 120 variables that include: income sources and amounts (yatra-derived and supplementary), household composition, productive assets, savings and debt, previous exposure to income shocks, awareness of the ropeway project, perceived

post-ropeway opportunities, and a 1–5 Likert scale stance on the ropeway project (1 = Strongly Oppose, 5 = Strongly Support).

Table 1: Sample composition by occupational group

Occupational group	N	Share (%)
Hospitality (hotel, lodge, dhaba, shop)	22	47.8
Animal handlers (pony, mule, palki)	12	26.1
Service (guide, priest, other)	6	13.0
Labour (porter/kandi)	3	6.5
Transport (driver)	3	6.5
Total	46	100.0

Hospitality workers form the largest group, reflecting their dominance in the visible Kedarnath economy. Animal handlers form the second-largest group and are, as the analysis shows, the most economically exposed to ropeway displacement.

3. Variable Construction and Livelihood Indices

3.1 Income Variables

The primary income variable is `total_income`: the sum of all reported annual income sources. Its natural logarithm (`log_income`) is used in regression specifications. The variable `share_yatra` captures what fraction of total income originates from yatra-season work. Income from agriculture, non-farm business, wage labour, remittances, government schemes, and animal husbandry are coded as zero where absent.

Eleven respondents had missing `dependency_ratio` values (the ratio of household dependents to earners), which were imputed with the column median. All other core variables were complete after coding.

3.2 Composite Livelihood Indices

Five composite indices were constructed following the Sustainable Livelihoods framework (DFID, 1999):

- **Vulnerability:** exposure to negative income shocks, proxied by a composite of shock history, debt levels, and low asset cover.
- **Adaptive capacity:** the ability to switch livelihoods or adjust household strategy in response to shocks. Reflects education, awareness of alternatives, and previous work history outside the yatra economy.

- **Asset index:** ownership of productive assets (land, animals, business premises) and their estimated resale value.
- **Diversification:** number of distinct income sources, reflecting resilience against demand shocks concentrated in a single activity.
- **Tourism dependence:** the degree to which the household’s economic participation is tied to pilgrim-facing activities, as opposed to activities that could persist regardless of pilgrim footfall.

These indices are used in regression models and principal components analysis. They are standardised before use in multivariate analysis.

4. Descriptive Analysis

4.1 Income Distribution

Figure 1 shows the distribution of annual income by occupational group. Hospitality and transport workers have the highest median incomes, while animal handlers are at the lower end of the distribution. The income composition panel (right) reveals that yatra-derived income accounts for over 95 % of mean income in every group. The pilgrimage economy is structurally mono-dependent: households have essentially no off-season income fallback that would buffer a decline in trail demand.

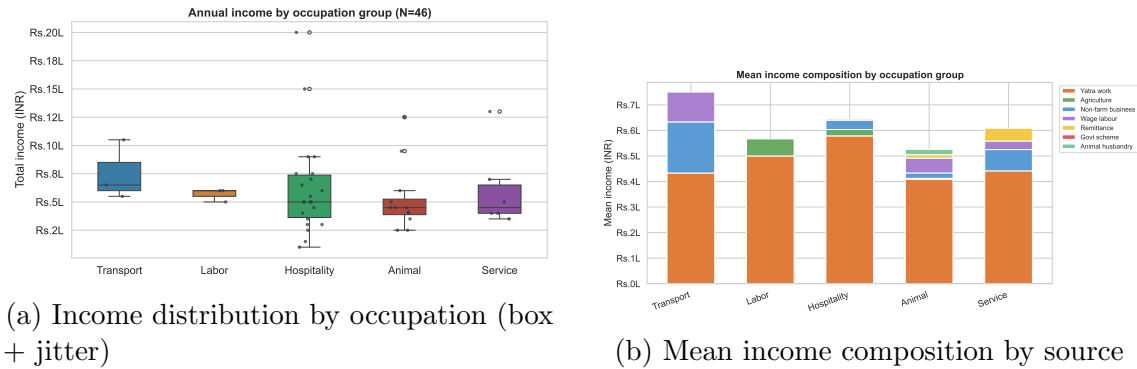


Figure 1: Annual income by occupational group ($N = 46$). Yatra work (orange in right panel) dominates every group’s income, leaving minimal off-season buffer.

4.2 Ropeway Stance Distribution

Table 2 presents the cross-tabulation of ropeway stance by occupational group. Of the 42 non-neutral respondents, 23 oppose the ropeway (stance 1 or 2) and 19 support it (stance 4 or 5). Attitudes are strongly patterned by occupation: animal handlers predominantly oppose (9 of 12, or 75 %), while hospitality workers predominantly support (10 of 22, or 45 %, with 9 of those at the highest support level). The

correlation heatmap (Figure 6) confirms that `support_ord` is negatively correlated with vulnerability ($r = -0.39$).

Table 2: Ropeway stance by occupational group (actual survey counts, $N = 46$)

Group	Likert scale (1 = Strongly Oppose, 5 = Strongly Support)					Total
	1	2	3	4	5	
Animal	9	0	0	0	3	12
Labour	1	1	0	0	1	3
Transport	0	1	0	1	1	3
Service	1	1	1	0	3	6
Hospitality	6	3	3	1	9	22
Total	17	6	4	2	17	46

Note: Oppose = stance 1 or 2 ($N=23$); Support = stance 4 or 5 ($N=19$); Neutral = stance 3 ($N=4$).

Animal handlers oppose at 75 % (9/12); hospitality workers support at 45 % (10/22).

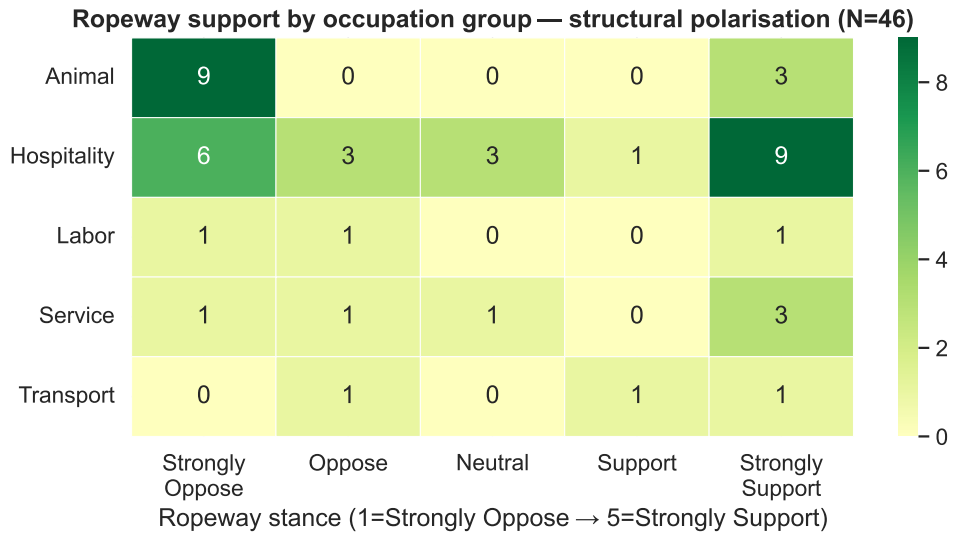


Figure 2: Ropeway stance heatmap by occupational group. The animal handler group is predominantly opposed while hospitality workers are predominantly supportive. The polarisation is occupationally structured.

4.3 Livelihood Profiles: Opposers versus Supporters

Figure 3a shows the standardised mean livelihood profiles of the 23 opposers and 19 supporters. Each variable is standardised to mean zero and standard deviation one so that indices measured on different scales can be compared on the same radar plot. The seven dimensions shown are: vulnerability, adaptive capacity, asset index, yatra income share, number of income sources, dependency ratio, and tourism dependence.

Opposers systematically score higher on vulnerability, yatra share, and dependency ratio, and lower on adaptive capacity, asset index, and number of income sources. This pattern indicates that those who oppose the ropeway have both more to lose (greater dependence on the trail economy) and less capacity to absorb or redirect that loss (lower adaptive capacity and fewer income sources). The pattern mirrors the scatter plot in Figure 3b, where respondents in the lower-left quadrant (below-median income, above-median vulnerability) are predominantly those with the strongest opposition.

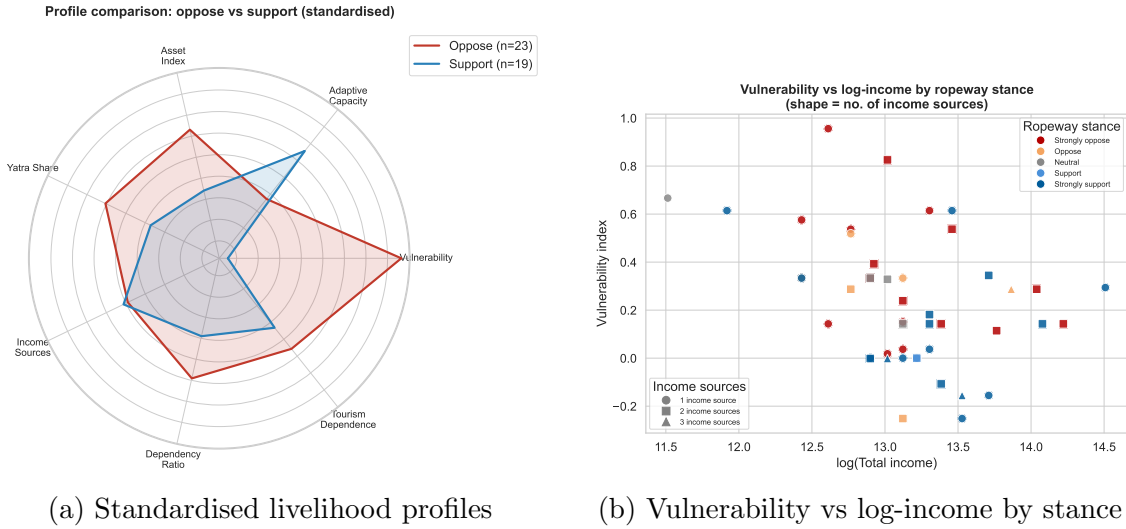


Figure 3: Left: standardised mean profiles of opposers vs supporters across seven livelihood dimensions. Right: scatter of vulnerability vs log-income, coloured by Likert stance. High-vulnerability, low-income respondents cluster toward strong opposition.

4.4 BPL Status and Financial Vulnerability

Figure 4 examines financial position by Below Poverty Line (BPL) card status. Among the 19 BPL card-holding respondents (41 % of the sample), savings rates are lower and vulnerability index values are higher than among the 27 non-BPL respondents. Income distributions for BPL households are concentrated at lower values. This illustrates a pre-existing stratum of acute financial fragility that a ropeway-induced income shock would be disproportionately likely to affect. Note that BPL card status is an imperfect proxy for income poverty in this sample — several card holders report incomes above the formal threshold — which is why occupational targeting is preferred over BPL-based targeting (Section 9).

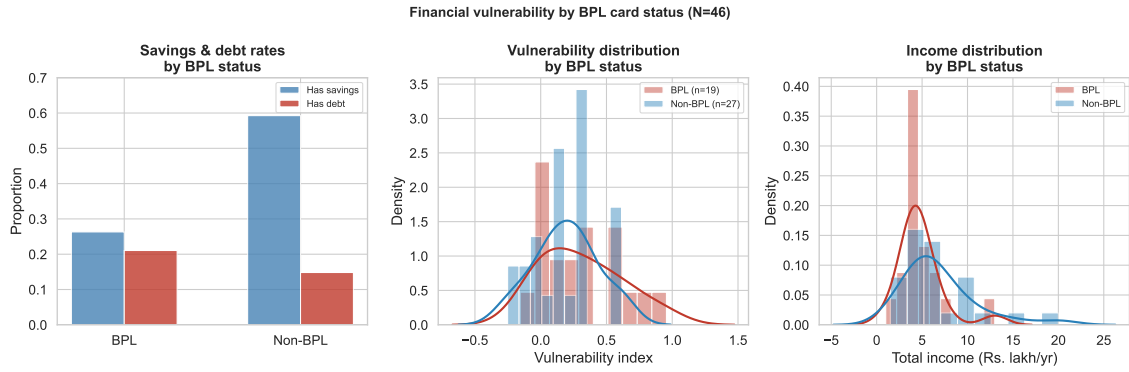


Figure 4: Financial position by BPL card status ($N = 46$). Left: proportion with savings or debt, by BPL status. Centre: vulnerability index distribution. Right: income distribution. BPL households show lower savings rates, higher vulnerability, and incomes concentrated at the lower end of the distribution.

4.5 Post-Ropeway Opportunity Perception

Figure 5 shows that 30 % of respondents (14 of 46) perceive no post-ropeway opportunities for themselves. Animal handlers and labourers are disproportionately concentrated in this category. Those who do perceive opportunities tend to identify hospitality or station-shop roles that are closely aligned with their existing activity. The animal handler group, whose current skill set is specific to trail-based animal transport, has the fewest perceived pathways into ropeway-adjacent employment — a finding that informs the design of the retraining scenario in Section 8.

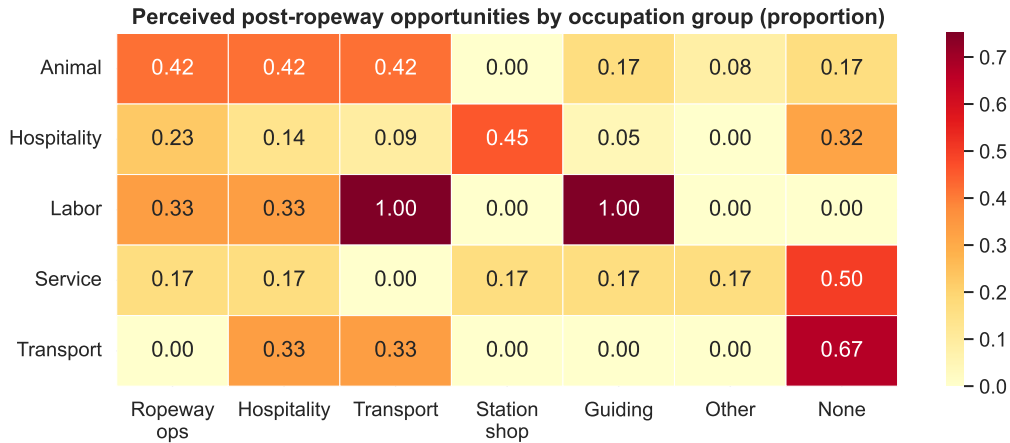


Figure 5: Proportion of each occupational group perceiving each type of post-ropeway opportunity. The “None” column indicates respondents who see no available opportunity for themselves. Animal handlers are heavily concentrated in this column.

4.6 Correlation Structure

The correlation matrix (Figure 6) reinforces a central finding of this study: workers who are more dependent on the trail are less able to cope with its loss. Income

sources, livelihood diversification, and adaptive capacity cluster tightly together ($r \approx 0.65\text{--}0.75$), and all three are negatively correlated with yatra income share and tourism dependence. Workers with no off-season alternatives are also the workers with fewest assets and least flexibility. The ropeway stance variable sits firmly in this structure: opposition is strongest among the most dependent and most vulnerable workers ($r = -0.39$ with vulnerability).

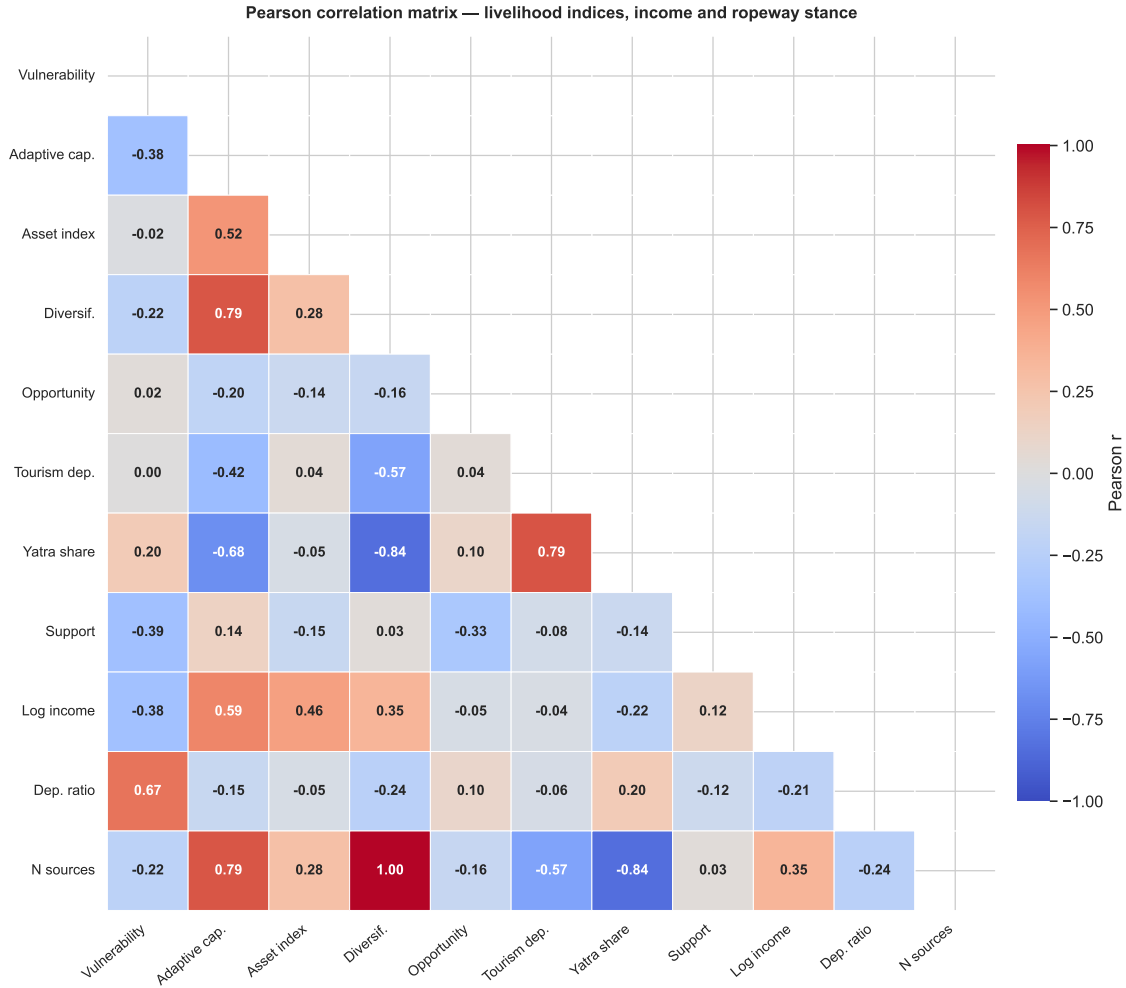


Figure 6: Pearson correlation matrix of livelihood indices, income, and ropeway stance (r , lower triangle). A resilience cluster (adaptive–diversification–income sources) and a dependence cluster (yatra share–tourism dependence) are the two main structural axes.

5. Statistical Models

5.1 OLS Income Model

What determines how much a worker earns? The clearest answer from the data is diversification: each additional income source a household holds is associated with a 35 % higher total income. Occupation group also matters in the expected direction

(animal handlers earn less than hospitality workers), but the signal is swamped by variation within groups at this sample size. A log-linear OLS regression with robust standard errors (HC3, to account for high-leverage hotel and lodge operators visible in Figure 7b) confirms this pattern formally ($R^2 = 0.183$, $N = 46$, income-sources coefficient $p = 0.013$). The takeaway for policy is straightforward: workers with no income alternatives are at the bottom of the earnings distribution and the most exposed to a demand shock. Equation 1 shows the full model specification.

$$\ln(\text{income}_i) = \alpha + \sum_{g \neq \text{Hosp.}} \beta_g \mathbb{I}[\text{group} = g]_i + \gamma_1 \cdot \text{n_sources}_i + \gamma_2 \cdot \text{dep_ratio}_i + \varepsilon_i \quad (1)$$

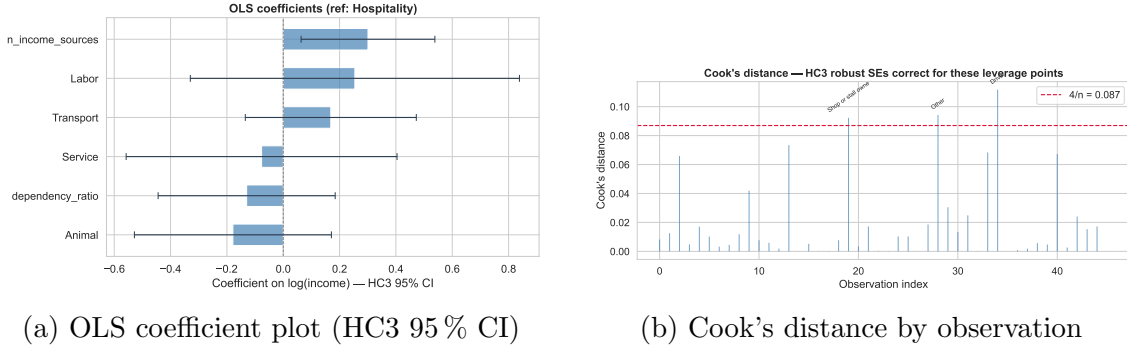


Figure 7: Left: OLS coefficient plot with HC3 confidence intervals; reference category is Hospitality. Right: Cook's distance; the dashed line marks the $4/N = 0.087$ threshold. High-leverage hotel and lodge owners inflate standard errors without HC3 correction.

Quantile regression at $q \in \{0.25, 0.50, 0.75\}$ (Figure 8) checks whether occupation-group effects differ across the income distribution. Divergence between quantile and OLS estimates indicates that aggregate coefficients mask heterogeneous effects at the tails — relevant because displacement policy should be designed for the most-exposed households, not the average.

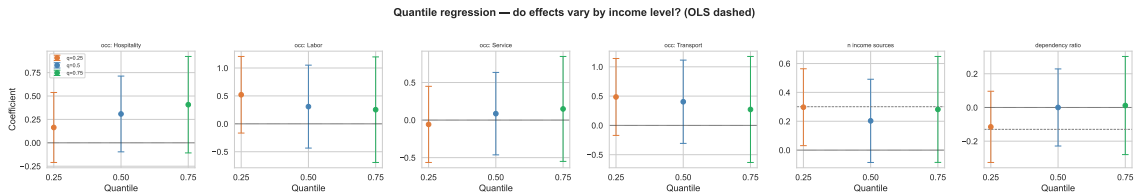


Figure 8: Quantile regression coefficients at $q = 0.25, 0.50, 0.75$ (coloured markers) versus the OLS estimate (dashed grey line). Divergence from OLS indicates that income effects are not uniform across the distribution.

5.2 Logit Model: Predicting Ropeway Opposition

Opposition is driven by vulnerability, not ignorance or occupation alone.

A logit model on the 42 non-neutral respondents finds that the single statistically significant predictor of opposition is the vulnerability index: a worker who scores one unit higher on the vulnerability scale is 60 percentage points more likely to oppose the ropeway, holding other factors constant. This is a structural finding. Workers oppose because they have much to lose and few alternatives — not because they misunderstand the project.

Being an animal handler increases the probability of opposition by 20 percentage points over an otherwise similar hospitality worker, but this effect does not reach significance ($p = 0.30$) because three animal handlers — those with lower vulnerability and higher asset scores — strongly support the project. Yatra income share, while conceptually important, adds no independent predictive power once vulnerability is controlled, which suggests that dependence on the trail matters primarily through its effect on household fragility.

Table 3 reports full coefficient estimates. The formal model specification (Equation 2) models the probability of opposing the ropeway as a logistic function of occupational group, vulnerability, and yatra-income share.

$$\Pr(\text{oppose}_i = 1) = \Lambda \left(\alpha + \sum_{g \neq \text{Hosp.}} \delta_g \mathbb{I}[\text{group} = g]_i + \phi_1 \cdot \text{vuln}_i + \phi_2 \cdot \text{share_yatra}_i \right) \quad (2)$$

Table 3: Logit model: probability of ropeway opposition ($N = 42$)

Predictor	Coef.	Std. Err. (HC3)	Marginal effect	p -value
Animal (vs Hospitality)	+1.022	0.979	+0.202	0.297
Labour	+0.302	1.242	+0.060	0.808
Service	−0.245	1.011	−0.048	0.808
Transport	−0.043	1.268	−0.008	0.973
Vulnerability index	+3.061	1.692	+0.604*	0.070
Share yatra	+0.716	1.948	+0.141	0.713
Intercept	−1.328	1.907	—	0.486
McFadden pseudo- R^2	0.160			
LLR p -value	0.160			
N	42			

Note: * Marginal effect significant at $p < 0.05$ level (dydx at overall means, z-stat = 2.15, $p = 0.031$). Vulnerability is the only predictor reaching conventional significance. Small N limits power across all estimates.

5.3 Logit Model: Probability of Having Savings

A second logit model asks who has financial reserves to weather the transition. Higher income is associated with more savings, while having a BPL card and a high dependency ratio both reduce the probability of holding savings. This creates a compounding disadvantage: the workers most exposed to the ropeway income shock are also the workers with the thinnest financial cushion to survive it (Equation 3).

$$\Pr(\text{savings}_i = 1) = \Lambda(\alpha + \beta_1 \ln \text{income}_i + \beta_2 \cdot \text{BPL}_i + \beta_3 \cdot \text{dep_ratio}_i) \quad (3)$$

5.4 Classification Models (LOOCV)

Can we predict which worker will oppose the ropeway from livelihood data alone? Three classifiers are tested using leave-one-out cross-validation (LOOCV), the appropriate method when $N < 50$ (Table 4). The best model (Random Forest) achieves 61.9 % accuracy against a 54.8 % baseline of always predicting opposition — a meaningful but modest improvement. The gap is not surprising: three animal handlers strongly support the ropeway, and six hospitality workers oppose it, making perfect separation impossible from occupation and vulnerability scores alone. The variables most predictive of held-out cases are yatra income share, the vulnerability index, and asset index (Figure 9), consistent with the logit results.

Table 4: Classification model performance — leave-one-out cross-validation

Model	LOO Accuracy
Logistic Regression ($C = 1.0$, L2 penalty)	0.571
Logistic Regression ($C = 0.1$, ridge)	0.571
Random Forest (depth 3, 300 trees)	0.619
Majority-class baseline (always predict oppose)	0.548

Note: All models outperform the majority-class baseline. AUC is not reported because LOO folds contain exactly one test sample.

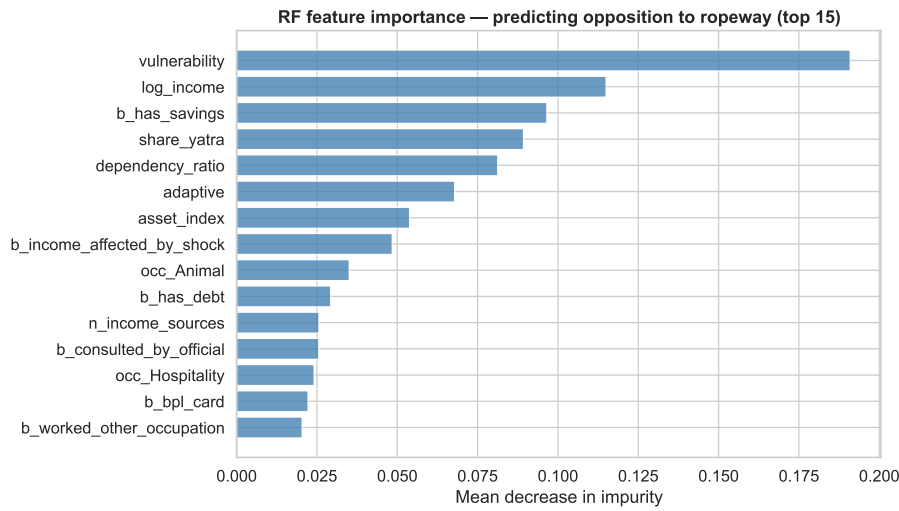


Figure 9: Random Forest feature importances (mean decrease in impurity, top 15 features). Yatra income share, vulnerability index, and asset index rank as the strongest predictors of opposition. Occupational dummies contribute additional signal.

6. Principal Components and Correspondence Analysis

6.1 PCA: A Resilience–Dependence Divide

The nine livelihood variables do not vary independently — they organise around two opposing poles. The first principal component, which explains almost half (46.8%) of all variation in the data, contrasts workers who have many income sources, high adaptive capacity, and low trail dependence against workers who are concentrated in a single activity, have few alternatives, and derive almost all income from the yatra season. This *resilience–dependence axis* is the dominant structural divide in the Kedarnath pilgrimage economy.

Animal handlers and labourers sit firmly at the dependence end of this axis (Figure 10). Their livelihood profile is a bundle of correlated disadvantages: high trail dependence,

narrow income base, low adaptive capacity, and high vulnerability — the same cluster that predicts ropeway opposition in the logit model. Hospitality workers are distributed more broadly, consistent with the within-group variation in their ropeway attitudes.

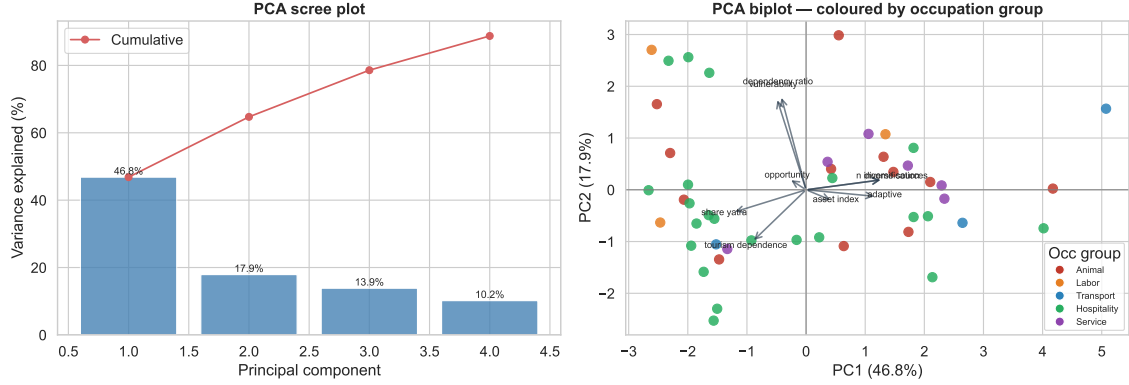
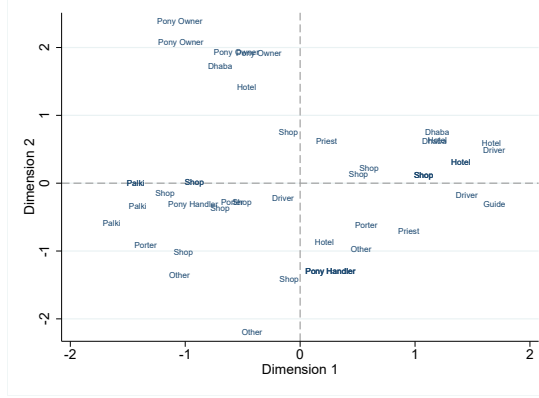


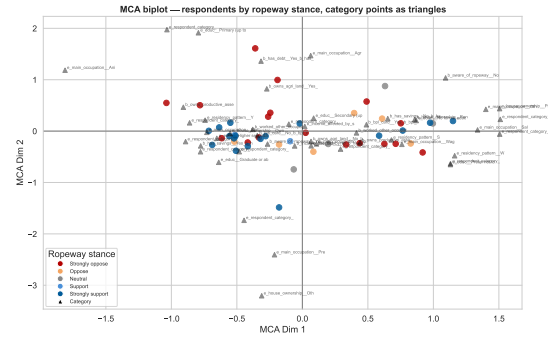
Figure 10: PCA scree plot (left) and biplot (right). PC1 (46.8% of variance) contrasts the resilience cluster (income sources, diversification, adaptive — positive loadings) against the trail-dependence cluster (yatra share, tourism dependence — negative loadings). Points are coloured by occupation group; arrows show variable loadings. Animal handlers and labourers cluster in the negative-PC1 region.

6.2 Multiple Correspondence Analysis

The MCA biplot (Figure 11) maps every worker and every categorical characteristic into the same space. The visual result is striking: strong opposition (darkest red) clusters around the same set of characteristics — BPL card, no savings, no productive assets, no work history outside the yatra economy. Strong support (darkest blue) clusters around ownership, education, and off-season diversification. This is not a model result; it is a geometric picture of the raw data, and it tells the same story as every other method used in this analysis: opposition to the ropeway is structurally rooted in economic fragility, not attitude or information.



(a) Stata MCA (support stance active variable)



(b) Python MCA (support stance as colour only)

Figure 11: Multiple Correspondence Analysis biplots. *Left*: Stata implementation with `support_ord` as an active variable; Dimension 1 is partly defined by the stance gradient. *Right*: Python (`prince`) implementation with stance used only for colouring; axes are defined independently of stance, providing a check that the same clusters emerge without stance driving the geometry. In both plots, strong opposition clusters with BPL, no savings, and no productive assets; strong support clusters with ownership and education categories.

7. Synthetic Population Generation and Validation

7.1 Why a Synthetic Population?

The 46-respondent survey is sufficient for understanding who these workers are and what drives their attitudes. But simulating the income effect of a ropeway on 12 animal handlers would produce estimates dominated by a handful of individuals. To generate stable group-level estimates for the scenario analysis, a synthetic population of 1,000 households is generated from the observed sample.

7.2 How It Was Built

Two methods are combined, each producing 500 households:

Smoothed bootstrap: Existing respondents are re-sampled with replacement, and continuous income and index variables receive a small random perturbation (Silverman bandwidth). Binary characteristics such as BPL card or savings status are copied without jitter, preserving discrete structure.

Gaussian copula: A statistical model is fitted to the joint distribution of all variables, capturing how they co-vary. New households are drawn from this model, preserving the resilience-dependence structure identified by PCA. This ensures the synthetic population inherits the same relationships between income, vulnerability, and occupation group that are present in the survey data.

Together, the two methods produce 1,000 synthetic households that replicate the survey’s distributional and correlation properties without repeating any individual respondent’s data.

7.3 Validation

The synthetic population closely matches the survey it was drawn from. Means, proportions, and group compositions all fall within 15 % of the observed values, and the correlation structure between livelihood variables is well-preserved (Frobenius norm of the correlation difference matrix = 0.596, versus an acceptable ceiling of 2.0). Four of 46 statistical tests flag minor distributional mismatches in discrete-heavy variables (yatra share, dependency ratio, asset index, diversification), where the continuous copula slightly smooths what are effectively step-function distributions in the small sample. All 12 categorical and binary checks pass. The overall verdict is **ACCEPT**: 42 of 46 checks pass (9 % failure rate, well inside the 20 % threshold). Table 5 reports the full statistics.

Table 5: Synthetic population validation: marginal statistics and KS tests

Variable	Obs. mean	Syn. mean	Rel. diff (%)	KS p	Result
log(income)	13.152	13.170	+0.1	0.580	PASS
Yatra share	0.838	0.823	−1.9	0.009	FAIL
Vulnerability	0.240	0.232	−3.5	0.640	PASS
Adaptive cap.	0.196	0.221	+13.1	0.168	PASS
Tourism dep.	0.298	0.283	−5.3	0.064	PASS
Dep. ratio	0.783	0.851	+8.7	0.002	FAIL
BPL card (prop.)	0.413	0.424	+2.7	—	PASS
Has savings	0.457	0.485	+6.2	—	PASS
Support (mean)	2.913	2.848	−2.2	—	PASS
Frobenius norm (corr. diff.)	0.596 (threshold: 2.0) — PASS				
TVD (support distribution)	0.076 (0 = identical)				
Overall	42/46 checks pass (9 % fail rate) — ACCEPT				

Note: KS = two-sample Kolmogorov–Smirnov test. Binary variables use a z -test for proportion equality. Rel. diff = $(\text{syn} - \text{obs})/\text{obs} \times 100$. TVD = total variation distance on the support-stance distribution.

Figures 12a and 12b show distribution overlays and category comparisons.

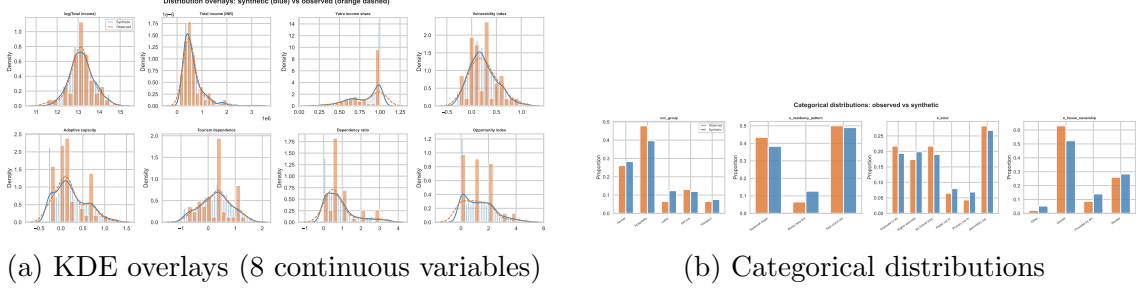


Figure 12: Synthetic (blue) vs observed (orange) distributions. KDE overlays confirm close marginal fit for continuous variables. Categorical proportions (occupation group, residency, education, housing) closely match the observed sample.

8. Scenario Analysis

8.1 Scenario Design

The ropeway’s economic impact is operationalised as a fractional reduction (or increase) in the yatra-derived share of household income. Five scenarios are simulated on the 1,000-household synthetic population:

- **Baseline:** No ropeway; all incomes unchanged.
- **Mild displacement (induced demand):** The ropeway increases total pilgrim throughput by approximately 40 %, partially offsetting trail-worker losses. Animal handlers and labourers suffer smaller income reductions; transport, service, and hospitality workers benefit.
- **Moderate displacement:** The ropeway operates but trail traffic is not entirely eliminated.
- **Severe displacement:** The ropeway captures almost all pilgrim movement; trail traffic falls sharply.
- **Ropeway with retraining:** Moderate displacement, but animal handlers and labourers gain 30 % of income through retraining and redeployment into ropeway-support roles.

Income shocks are applied to the yatra-derived income portion only:

$$y_i^{\text{post}} = y_i^{\text{yatra}} \cdot (1 + \delta_{g(i)}) + y_i^{\text{other}} \quad (4)$$

where $\delta_{g(i)}$ is the group-specific shock for occupation group g , $y_i^{\text{yatra}} = \text{share_yatra}_i \times y_i^{\text{pre}}$, and y_i^{other} is the non-yatra income. Table 6 shows the shock parameters.

Table 6: Income shock parameters (δ_g) by scenario and occupational group

Group	Mild	Moderate	Severe	+ Retraining
Animal	−40 %	−75 %	−95 %	−75 % + 30 % gain
Labour	−30 %	−60 %	−80 %	−60 % + 30 % gain
Transport	+8 %	−25 %	−40 %	−25 %
Service	+12 %	+5 %	+5 %	+5 %
Hospitality	+35 %	+20 %	+25 %	+20 %

Note: Shocks are applied to yatra-derived income only. **Mild scenario:** ropeway increases total pilgrim throughput $\sim 40\%$, partially offsetting trail-worker losses; gains for Transport/Service/Hospitality are applied to total income (increased footfall). Effective income changes are smaller than shock parameters because mean `share_yatra` < 1 for most groups. Retraining gain is an additional fractional increase in total income for Animal and Labour households.

8.2 Group-Level Income Impacts

Table 7 shows mean income by group before and after each scenario, along with the effective income change. Effective changes are smaller than the shock parameters because mean `share_yatra` in the synthetic population is below 1.0 for most groups.

Table 7: Mean income impact by occupational group across scenarios (Rs. lakh/year)

Group	Baseline	Mild	Moderate	Severe	+ Retraining
Animal	5.22	3.55 (−32%)	2.09 (−62%)	1.25 (−78%)	3.66 (−32%)
Labour	6.38	4.87 (−24%)	3.35 (−50%)	2.34 (−66%)	5.26 (−18%)
Transport	7.74	8.36 (+8%)	6.47 (−18%)	5.71 (−26%)	6.47 (−18%)
Service	6.62	7.41 (+12%)	6.38 (−4%)	6.38 (−4%)	6.38 (−4%)
Hospitality	6.45	8.71 (+35%)	7.74 (+20%)	8.07 (+25%)	7.74 (+20%)

Note: Income in Rs. lakh/year (1 lakh = Rs. 1,00,000). Percentage changes in parentheses are effective changes, which are smaller than shock parameters because `share_yatra` < 1 for most households. Mild scenario: animal handlers (−32%) and labourers (−24%) face reduced but non-negligible losses; transport, service, and hospitality gain from induced pilgrim demand. Source: `ropeway.scenarios`.

Figure 13a visualises these impacts; Figure 13b shows them as a heatmap.

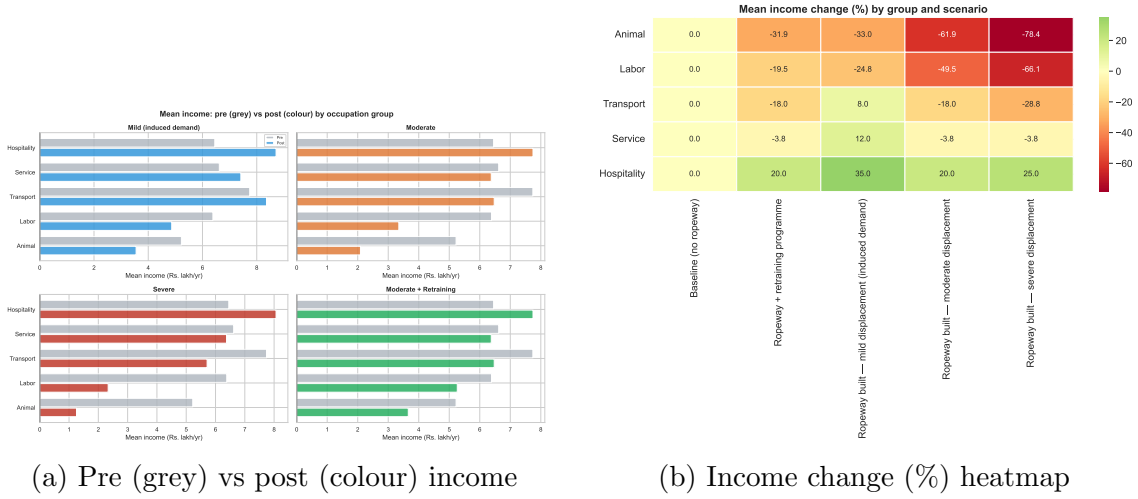


Figure 13: Scenario impacts on mean income by occupational group (four-panel, two per row). Mild scenario: ropeway-induced pilgrim growth limits losses for animal handlers and labourers while boosting transport, service, and hospitality. Retraining narrows losses further for the most affected groups.

8.3 Poverty Headcount

A monthly poverty line of Rs. 3,500 (Tendulkar Committee threshold, inflation-adjusted; Planning Commission of India 2009) is used to count households below the poverty threshold pre- and post-scenario. Table 8 and Figure 14 summarise the results.

Table 8: Poverty headcount by scenario ($N = 1,000$ synthetic households; poverty line: Rs. 3,500/month)

Scenario	N below poverty	% below	Newly poor
Baseline (no ropeway)	0	0.0	0
Mild (induced demand)	0	0.0	0
Moderate displacement	3	0.3	3
Severe displacement	121	12.1	121
Ropeway + retraining	0	0.0	0

Note: Newly poor = households above the poverty line at baseline that fall below it post-scenario.

All 121 newly poor in the severe scenario are animal handlers ($n_{\text{Animal}} = 283$ in synthetic population; approximately 42% of the group fall below the poverty line). The mild scenario avoids poverty induction entirely, as induced pilgrim growth offsets the income losses for trail-dependent workers.

The mild scenario avoids poverty induction entirely: increased pilgrim throughput boosts incomes for transport, service, and hospitality workers, and limits the loss for animal handlers (effective -32%) and labourers (-24%) to levels well above the poverty threshold. Under the moderate scenario, very few synthetic households cross

the poverty threshold because mean animal handler income after a -62% effective reduction still exceeds Rs. 42,000/year (post-shock mean: Rs. 2.09 lakh). Under the severe scenario, however, 42 % of the synthetic animal handler group fall below the poverty line. Retraining eliminates poverty induction in the moderate case.

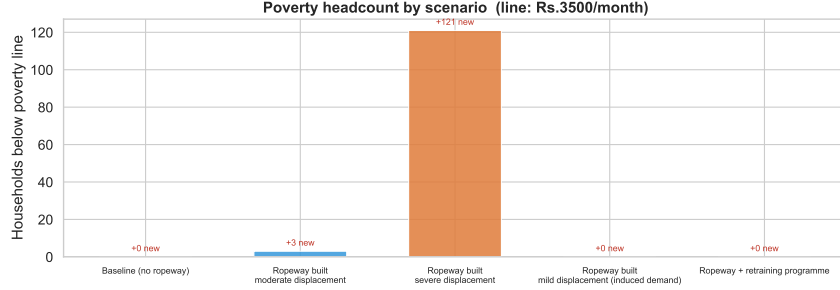


Figure 14: Poverty headcount by scenario. Numbers above bars indicate newly poor households (those above the threshold at baseline). The jump from moderate to severe is driven entirely by animal handlers.

8.4 Compensation Analysis

For each household with a net income loss, the minimum lump-sum compensation needed to restore a three-month income buffer is:

$$C_i = \max\left(0, 3 \times \frac{y_i^{\text{pre}} - y_i^{\text{post}}}{12}\right) \quad (5)$$

The three-month buffer is the policy target: it is sufficient to prevent distress asset sales or forced debt during a transition period but does not provide long-term income replacement.

Table 9 reports group-level compensation requirements for the moderate and severe scenarios.

Table 9: Minimum lump-sum compensation to restore 3-month income buffer

Group	Moderate scenario		Severe scenario	
	Mean/hh (Rs.)	Total (Rs. lakh)	Mean/hh (Rs.)	Total (Rs. lakh)
Animal	78,331	221.7	99,209	280.8
Labour	75,909	94.1	101,213	125.5
Transport	31,679	24.1	50,687	38.5
Service	6,089	7.3	6,089	7.3
Hospitality	0	0.0	0	0.0
Total	—	347.2	—	452.1

Note: All 1,000 synthetic households that have a net income loss are compensated ($N_{\text{compensated}} = 603$ moderate, $N = 603$ severe for losing groups). Hospitality workers gain income in both scenarios and receive no compensation. Totals: Rs. 3.47 crore (moderate) and Rs. 4.52 crore (severe).

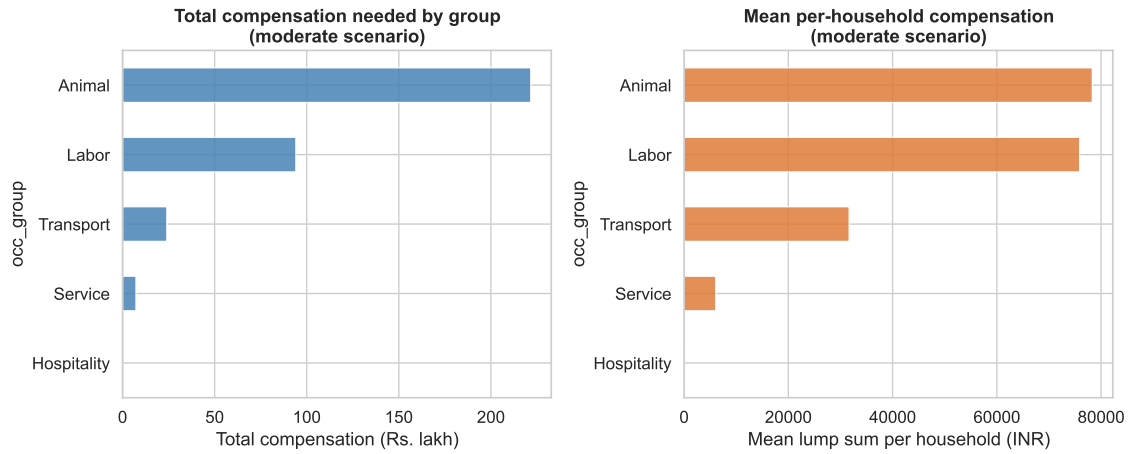


Figure 15: Compensation cost by group (moderate scenario). Left: total outlay per group. Right: mean per-household lump sum. Animal handlers require the largest total outlay owing to both their group size (283 households) and the magnitude of their income loss.

The total fiscal cost — Rs. 3.47 crore under moderate assumptions, Rs. 4.52 crore under severe — is small relative to the estimated ropeway project cost (several hundred crore). The aggregate cost is also dominated by animal handlers and labourers, making occupational targeting efficient: the same fiscal envelope, if redirected to all workers regardless of occupation, would be less cost-effective.

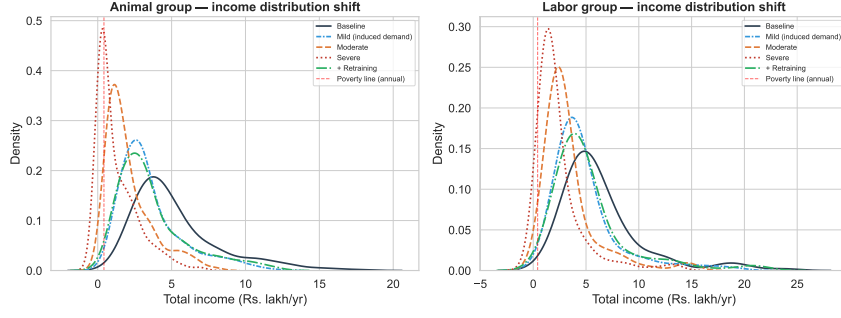


Figure 16: Income distribution of animal handlers (left) and labourers (right) across all five scenarios. The mild scenario shifts the distribution rightward (induced demand). The severe scenario shifts it substantially below the poverty line (red dashed). Retraining largely restores the baseline distribution for both groups.

9. Policy Implications

9.1 Occupational Targeting is More Efficient than BPL Targeting

The analysis shows that BPL card status is an imperfect proxy for ropeway exposure. The 19 BPL card-holding respondents overlap substantially with the most exposed groups (animal handlers and labourers), but the overlap is not complete. Compensation eligibility based on occupational registration — pony handler licence, porter registration, or equivalent documentation — would reach all structurally exposed workers regardless of their formal poverty classification. BPL-based targeting would miss a portion of the most-affected group.

9.2 Compensation Must Precede Displacement

The vulnerability data show that a significant share of animal handler households carry debt and hold limited savings. An income shock that arrives before compensation is disbursed would trigger distress asset disposal, debt spirals, or out-migration. These dynamics are difficult to reverse (Bryceson, 2000). The compensation formula in Equation 5 is designed as a pre-emptive buffer, not a post-facto relief payment. Practically, this means disbursement should be completed before the ropeway commences commercial operations, not after it has been running for a season.

9.3 Retraining Requires Demonstrated Placement, Not Just a Programme

The retraining scenario produces the most favourable outcome: all poverty induction is eliminated and animal handler income losses are halved. However, 30 % of survey respondents currently see no post-ropeway opportunities for themselves, and animal handlers are over-represented in this group. A retraining programme will not produce uptake if prospective participants do not believe placements are available. The design

implication is that placements in ropeway operations, station services, or eco-tourism guiding should be publicly committed and visible before the retraining programme is announced.

10. Conclusion

The Kedarnath ropeway will be a transformative improvement for pilgrims. For the workers who have built livelihoods around the foot trail, it is a threat — and not an equal one. This report shows that the threat falls hardest on animal handlers, whose entire economic life is organised around an activity the ropeway renders redundant. These workers are not opposed because they misunderstand the project. They are opposed because they have the most to lose and the fewest means to absorb that loss.

The scenario analysis translates this into numbers. Even in the optimistic mild scenario — where the ropeway draws more pilgrims and partly offsets trail-worker losses — animal handlers still lose about a third of their income. Under moderate displacement the loss is 62 %. Under the most severe assumptions, 42 % of animal handler households fall below the poverty line.

The good news is that prevention is cheap. A targeted scheme costing Rs. 3.5–4.5 crore — a fraction of a percent of the project budget — can restore a three-month income buffer for every displaced household and eliminate new poverty entirely. Retraining, if linked to visible job placements before the ropeway opens, can do the same at lower fiscal cost.

The central lesson is about sequencing. Compensation and retraining must come *before* displacement, not after. An income shock that arrives before the safety net is in place will trigger debt, distress asset sales, and out-migration that no subsequent payment can reverse. The technology is not the problem. The transition design is.

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